

Vishal Kumar Allam

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PROFESSIONAL SUMMARY

- QA Automation Engineer with over **6+ years** of experience designing scalable automated testing frameworks for enterprise applications, APIs, mobile platforms, connected systems, and hardware/software-integrated solutions in fast-paced engineering environments.
- Strong hands-on expertise using **Python, Java, C#, Selenium WebDriver, Playwright, Appium, RESTAssured, Cucumber BDD, and JMeter** to build reliable functional, integration, smoke, regression, end-to-end, and performance automation.
- Experienced evolving traditional and AI-assisted QA approaches toward **agentic testing and agentic engineering workflows**, using intelligent automation to generate scenarios, orchestrate execution, analyze failures, and improve coverage across product lines.
- Strong understanding of **firmware validation, embedded systems, IoT-style devices, hardware-software communication, device integrations, APIs, and internal applications**, with emphasis on repeatable testing across system boundaries.
- Hands-on experience integrating automated suites with **Azure DevOps, GitHub Actions, Jenkins, Git, and CI/CD pipelines**, enabling continuous validation of every build, automated quality gates, and faster release feedback.
- Experienced designing resilient automation architectures that reduce repetitive manual testing, improve regression reliability, support hardware and firmware updates, and shorten feature delivery without compromising product quality or security.
- Collaborative Quality Engineer experienced partnering with DevOps and Development teams, reviewing automation architecture, troubleshooting complex failures, evaluating new tools through POCs, and continuously improving testing efficiency and engineering standards.

EDUCATION

University of Missouri Kansas City

"Master of Science" in Computer Science

Kansas City, MO

Jan. 2023 – May 2024

Jawaharlal Nehru Technological University

"Bachelor of Technology" in Computer Science

Hyderabad, India

Aug. 2015 - May 2019

TECHNICAL SKILLS

- **Programming & Automation:** Python, Java, C#, JavaScript, TypeScript, SQL, Selenium WebDriver, Playwright, Appium, RESTAssured, Cucumber BDD, JUnit, TestNG, NUnit.
- **AI & Agentic Testing:** Agentic Testing, Agentic Engineering, AI-Assisted Test Generation, Autonomous Test Workflows, Intelligent Failure Analysis, Test Orchestration, AI-Based Coverage Expansion, Automation Optimization.
- **Embedded & Hardware Integration:** Firmware Testing, Embedded Systems, IoT Devices, Hardware-Software Communication, Device Integration, Access-Control Systems, Sensor/Controller Workflows, System Interface Validation.
- **Testing Practices:** Smoke Testing, Regression Testing, Integration Testing, End-to-End Testing, API Testing, Performance Testing, Functional Testing, Hardware/Firmware Validation, Continuous Validation.
- **API & Performance:** Postman, REST/JSON APIs, Swagger/OpenAPI, RESTAssured, JMeter, Load Testing, Performance Benchmarking, Backend Validation, Data Integrity, Service Integration Testing.
- **DevOps & CI/CD:** Azure DevOps, GitHub Actions, Jenkins, Git, GitHub, CI/CD Pipelines, Automated Quality Gates, Parallel Execution, Build Validation, Release Automation, Continuous Testing.
- **Engineering Practices:** Automation Framework Architecture, Reusable Test Components, Test Reliability, Root Cause Analysis, Tool Evaluation, POCs, DevOps Collaboration, Agile/Scrum, Code Reviews, Continuous Improvement.

Test Automation Engineer

PCMI LLC (Full Time)

Oct 2024 – Present

Falls Church, VA, USA

- Design and maintain scalable automation frameworks using **Python, C#, Playwright, Selenium WebDriver, REST APIs, and SQL**, supporting reusable smoke, integration, regression, and end-to-end validation across complex enterprise product workflows.
- Extend AI-assisted automation toward **agentic testing workflows** that generate scenarios, orchestrate test execution, analyze failures, and recommend targeted regression coverage while maintaining engineering review of autonomous test decisions.
- Develop system-level validation for hardware/software-integrated workflows by testing service communication, device-facing interfaces, backend APIs, configuration changes, and firmware-style update scenarios across interconnected components.
- Integrate automated test suites with **Azure DevOps, TeamCity, Git, and CI/CD pipelines**, enabling continuous validation, parallel regression execution, build quality gates, and faster detection of release-impacting defects.
- Build performance and reliability scenarios using JMeter, APIs, SQL, logs, and application telemetry to identify latency, throughput, synchronization, integration, and stability issues under representative workloads.
- Perform root-cause analysis of automation, application, environment, and integration failures using logs, traces, API responses, SQL diagnostics, and debugging while implementing durable fixes rather than repetitive workarounds.
- Partner with developers and DevOps engineers to evaluate new testing tools, build proof-of-concepts, review automation code, improve reusable framework architecture, and reduce manual effort across recurring release and maintenance cycles.

Environments: Python, C#, Playwright, Selenium WebDriver, REST APIs, Postman, SQL Server, JMeter, Azure DevOps, TeamCity, Git, CI/CD, Agentic Testing, AI-Assisted QA, Hardware/Software Integration, Continuous Validation, Root Cause Analysis, Agile/Scrum.

SDET / QA Automation Engineer

Client: CVS

Dec 2023 – Sept 2024

Falls Church, VA, USA

- Developed scalable automated validation using **Java, TypeScript, Playwright, Selenium WebDriver, Appium, RESTAssured, and SQL** across web, mobile, API, and distributed backend applications supporting high-volume healthcare workflows.
- Built reusable regression frameworks and AI-assisted testing utilities that accelerated scenario generation, defect investigation, test maintenance, and coverage analysis while improving automation reliability across frequently changing features.
- Validated connected application and device workflows through Appium/mobile automation, API testing, backend verification, environment configuration, and integration scenarios that exercised communication across multiple software components.
- Integrated automated suites with **Jenkins, GitLab/GitHub, Git, and CI/CD pipelines**, supporting smoke, regression, API, and performance execution as part of continuous build and release validation.
- Executed JMeter-based performance tests and API diagnostics to measure response times, throughput, concurrent processing, and service stability while identifying bottlenecks affecting customer-facing and integrated workflows.
- Investigated flaky tests, service failures, and integration defects through Splunk, application logs, API traces, SQL, and debugging techniques while partnering with developers on rapid root-cause analysis and resolution.
- Collaborated with DevOps, software engineers, product teams, and SDETs to improve automation architecture, review test code, evaluate emerging tools, and promote scalable, repeatable Quality Engineering practices.

Environments: Java, TypeScript, Playwright, Selenium WebDriver, Appium, RESTAssured, Postman, SQL, JMeter, Jenkins, GitLab/GitHub, Git, CI/CD, Splunk, AI-Assisted Testing, Mobile/Device Integration, Performance Testing, Agile/Scrum.

Software Engineer

Infor

Sept 2021 – Dec 2022

Hyderabad, India

- Designed and maintained automation solutions using **Java, Python, Selenium WebDriver, Cucumber BDD, JUnit, TestNG, REST APIs, and SQL**, covering functional, integration, regression, and system-level validation across enterprise applications.
- Built reusable framework components, test utilities, service abstractions, and execution helpers while applying object-oriented design principles to improve scalability, test reliability, and long-term maintainability.

- Validated hardware/software-style integration boundaries through service interfaces, device-oriented communication patterns, configuration dependencies, data exchange, and error scenarios across connected enterprise components.
- Integrated automated suites with **Jenkins, Git, Maven, and CI/CD pipelines**, enabling continuous smoke and regression execution, build-quality checks, repeatable testing, and earlier identification of integration defects.
- Performed performance and reliability testing using JMeter, service metrics, application logs, and backend diagnostics to evaluate throughput, response times, resource behavior, and stability across integrated systems.
- Troubleshoot application and automation failures using logs, SQL, API responses, debugging, and root-cause techniques while refactoring brittle test components and strengthening regression protection around recurring issues.
- Worked with development and QA teams to evaluate testing tools, prototype automation improvements, strengthen test architecture, and continuously improve engineering processes across Agile software releases.

Environments: Java, Python, Selenium WebDriver, Cucumber BDD, JUnit, TestNG, Postman, REST APIs, SQL, JMeter, Jenkins, Git, Maven, CI/CD, Integration Testing, Hardware/Software Interfaces, Framework Development, Root Cause Analysis, Agile/Scrum.

Associate Engineer QA

June 2019 – August 2021

Virtusa

Hyderabad, India

- Developed and executed automated testing using **Java, Selenium WebDriver, Appium, Cucumber BDD, JUnit, TestNG, REST APIs, and SQL** across web, mobile, API, and integrated enterprise applications.
- Created reusable functional, smoke, integration, regression, and end-to-end automation suites while reducing repetitive manual validation and expanding coverage of high-value business and system workflows.
- Validated mobile and connected-system behavior through Appium, physical/emulated devices, API integrations, configuration testing, backend data verification, and device-specific scenarios across supported application environments.
- Performed API and backend testing using **Postman, REST/SOAP services, JSON/XML, and SQL**, verifying service communication, data integrity, business rules, error handling, and cross-component integration behavior.
- Integrated regression suites with **Jenkins, Git, and CI/CD pipelines**, enabling automated build validation, repeatable smoke testing, regression execution, and faster feedback throughout iterative release cycles.
- Supported performance validation using JMeter and application diagnostics while troubleshooting automation and integration failures through logs, API responses, database queries, device evidence, and root-cause analysis.
- Collaborated with developers, analysts, and QA engineers to review automation approaches, improve reusable testing standards, identify coverage gaps, and continuously enhance test efficiency, reliability, and maintainability.

Environments: Java, Selenium WebDriver, Appium, Cucumber BDD, JUnit, TestNG, Postman, REST APIs, SQL, JMeter, Jenkins, Git, CI/CD, Mobile/Device Testing, Hardware-Software Integration, Smoke/Regression Testing, Root Cause Analysis, Agile/Scrum.

AWARDS & ACHIEVEMENTS

- 2019-11 Oracle Certified Associate, JAVA SE8 Programmer.
- ISTQB Certified Professional.
- Microsoft Technology Associate (MTA) – Database Administration Fundamentals, HTML- 5 Application Development Fundamentals.
- Awarded “Software Process Achievement” Award for my contribution towards the project in Virtusa.
- Received Customer Appreciation mails for quality releases.